Information Sciences 647 (2023) 119439

Contents lists available at ScienceDirect

Information Sciences

journal homepage: www.elsevier.com/locate/ins

A hybrid MCDM model with Monte Carlo simulation to improve decision-making stability and reliability

Haizhou Cui [a,1], Songwei Dong [b,1], Jiayi Hu [c,1], Mengqi Chen [d,1], Bodong Hou [e,2], Jingshun Zhang [f,2], Botong Zhang [g,2], Jitong Xian [h,2], Faan Chen [i,*]

[a] School of Economics, University of Sydney, Camperdown, New South Wales 2006, Australia
[b] School of Information Science and Technology, University of Science and Technology of China, Hefei, Anhui 230026, China
[c] Department of Mathematics, Columbia University, New York, NY 10027, United States
[d] Department of Electrical and Computer Engineering, New York University, New York, NY 11201, United States
[e] Department of Electrical and Computer Engineering, Northwestern University, Evanston, IL 60208, United States
[f] Department of Statistics, Columbia University, New York, NY 10027, United States
[g] Department of Computer and Information Science, University of Pennsylvania, Philadelphia, PA 19104, United States
[h] Department of Computer Science, Johns Hopkins University, Baltimore, MD 21218, United States
[i] School of Engineering and Applied Sciences, Harvard University, Cambridge, MA 02138, United States

ARTICLE INFO

Keywords:
CRITIC
MABAC
K-means
Monte Carlo simulation
Stability
Reliability

ABSTRACT

Employing an appropriate method to achieve a reliable decision remains a challenge for decision-makers (DMs) in the multiple-criteria decision-making (MCDM) process owing to its inherent model-related complexity, sensitivity, and uncertainty. In this context, this study proposes an innovative hybrid MCDM model that integrates criteria importance through intercriteria correlation (CRITIC), multi-attributive border approximation area comparison (MABAC), and k-means with Monte Carlo simulation (i.e., CRITIC–MABAC–Kmeans with Monte Carlo simulation), aiming to address MCDM problems with substantial stability and reliability. Specifically, MABAC attests to the stability of this method, as it is less affected by normalization and weighting schemes. In addition, the challenge of conflicting k-means clustering outcomes, owing to diverse initial centroid selections, is mitigated by a Monte Carlo simulation, which identifies the most probable type of result and compensates for small-sample size bias. The model performance is tested using a case study of observing transport safety accomplishments in the ASEAN region. Enhanced multiple comparisons of the experimental results verify the quality, efficiency, and adaptability of the proposed model, indicating its feasibility for DMs, policymakers, and practitioners as a practical tool for handling real-life MCDM activities in various domains under compounded sensitivity and uncertainty.

1. Introduction

Multiple criteria decision-making (MCDM) problems involve several alternatives evaluated on multiple indicators to help decision-

* Corresponding author at: Harvard University, Pierce Hall G2C, 29 Oxford Street, Cambridge, MA 02138, United States.
E-mail address: faanchen@seas.harvard.edu (F. Chen).
[1] Co-first authors of the article.
[2] Co-second authors of the article.

https://doi.org/10.1016/j.ins.2023.119439
Received 26 July 2022; Received in revised form 22 July 2023; Accepted 3 August 2023
Available online 12 August 2023
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makers (DMs) determine the best option. MCDM methods are designed to reduce subjectivity in decision-making by applying a series of filters to the selection process [1]. However, given the increasing complexity and ambiguity of real-world decision-making issues, DMs face increasing difficulty in offering accurate judgments [2]. MCDM methods are characterized by a particular mathematical apparatus that can result in different solutions when different methods are applied to the same problem [3]. This indicates that the final decision (alternative preference) is not only dependent on the chosen indicators but also sensitive to the selected MCDM method [4]. In other words, being capable of handling decision problems with high stability and reliability is critical for performing MCDM activities.

The MCDM process typically involves three steps [5]. First, a weight method is employed to determine the relative importance of each indicator, given their different levels of contribution to the performance of alternatives. This is because an equal-weighted approach may over-consider indicators with smaller impact factors on individual performance and under-consider those with larger impact factors [6]. Second, DMs often use an aggregating (ranking) method to determine the order of their preferences for alternatives. This is the key step during MCDM, as the final ranking result represents DMs' preferences and influences the final decision. However, previous studies have shown that variation in the weights of indicators may lead to different final solutions from the same ranking method [7,8]. This indicates the need for a stable aggregating method that is less prone to disruption by variations in weight selection. Third, many MCDM applications require a process of clustering alternatives based on their ranks for DMs to manage alternatives categorically [9,10]. In addition, with the development of artificial intelligence, data-driven approaches with machine learning have been widely used in a variety of ways to increase decision-making quality and efficiency. Previous studies have laid a solid foundation for solving real-life decision-making problems in various domains. Nevertheless, considerable sensitivity and uncertainty exist in the decision process owing to the variety of MCDM methods available, particularly for small-sample scenarios or cases, which remain a challenge for DMs. In this context, a comprehensive MCDM approach that can combine the three steps (weighting, aggregating, and grouping) into an all-in-one framework with good stability and reliability should be identified.

To this end, we develop a brand-new hybrid MCDM model with Monte Carlo simulation that integrates criteria importance through intercriteria correlation (CRITIC), multi-attributive border approximation area comparison (MABAC), and k-means with Monte Carlo simulation, namely CRITIC–MABAC–Kmeans with Monte Carlo simulation, to address the challenge that previous MCDM models encountered in decision-making surroundings under model-related sensitivity and uncertainty, providing DMs with a decision support system with stability, reliability, and efficiency. The proposed hybrid MCDM model is illustrated and tested through an application of transport safety accomplishments observed in the ASEAN region. This novel study contributes to academia and society in the following ways: (1) The proposed model seamlessly connects weighting, aggregating, and grouping and embeds a Monte Carlo simulation. This improves the stability, reliability, and applicability of the decision-making process and enriches the existing MCDM system. (2) It serves as a prototype to address bias and result discordance caused by different initial centroid types in the k-means algorithm through a Monte Carlo simulation. This can be seen as a precursor to utilizing the Monte Carlo simulation nested k-means algorithm in small-sample scenarios. (3) This study provides governmental officials, DMs, and professionals of ASEAN countries, as well as other countries worldwide, with a promising tool to effectively monitor governance performance (e.g., transport safety accomplishments) and enhance their capabilities in management, decision-making, and policymaking.

2. Literature review

2.1. *Weighting*

Weighting assignment is a key element in the MCDM process and plays an important role in measuring the overall performance (i.e., composite index) of alternatives and their corresponding rankings [11]. Typically, weighting-determination methods can be categorized into subjective and objective methods [6].

Subjective weight determination is usually performed using a certain rule based on the subjective experience and judgment of DMs or experts, who are generally provided with a set of questions that allow them to directly express their opinions on the relative importance of the criteria. The most popular subjective weighting methods include the analytic hierarchy process (AHP), analytic network process (ANP), conjoint analysis (CA), budget allocation process (BAP), Delphi method (also known as estimate-talk-estimate [ETE]), fixed point allocation (FPA), direct rating, swing weighting, and simple multi-attribute rating technique (SMART).

Objective weighting methods generally obtain weights through mathematical and statistical models, and DMs play no role in determining the relative importance of the criteria. The popular objective methods include the entropy method, data envelopment analysis (DEA), factor analysis (FA), principal component analysis (PCA), coefficient of variation (CV), independent weighting (IW), preference selection index (PSI), CRITIC, standard deviation (SD), and objective-subjective weighted method for minimizing inconsistency (OSWMI). Additionally, equal weighting (EW) is a widely used method for creating a composite index in the MCDM process (e.g., Human Development Index, Environmental Sustainability Index, and Technology Achievement Index), which assigns weights equally to all criteria.

Overall, both subjective and objective weighting methods have advantages and disadvantages. Utilizing a subjective weighing approach disregards decision information and overly relies on expert preferences, compromising the objectivity of the results [12]. Besides, subjective weighting methods are often time-consuming, especially in situations where experts do not agree on the considered problem and/or the number of criteria is large [13]. However, they generally include opinions from various stakeholders (e.g., experts, citizens, and politicians) and explain the elicitation process more clearly. This approach can better reflect policy priorities and theoretical factors [11]. Most importantly, subjective weighting methods are not limited by restrictions. In contrast, objective methods require constructing a numerical decision matrix and entail additional mathematical skills and effort [14]. However, objective weighting methods can achieve significant advantages in terms of computational efficiency when there are several criteria.

2.2. Aggregating

Aggregating combines an alternative's performance across numerous qualitative and/or quantitative criteria and leads to a solution that requires consensus [15]. Determining an appropriate aggregating method to combine multidimensional variables into a composite index is a crucial step in the MCDM process.

Several aggregating methods are available in the MCDM process [16,17], including DEA, AHP, technique for order preference by similarity to an ideal solution (TOPSIS), *vise kriterijumska optimizacija i kompromisno resenje* (VIKOR; translated from Serbian as "multicriteria optimization and compromise solution"), rank-sum ratio (RSR), Condorcet-Kemeny-Young-Levenglick ranking procedure (C-K-Y-L), elimination and et choice translating reality (ELECTRE), interactive multi-criteria decision-making (TODIM), preference ranking organization method for enrichment evaluation (PROMETHEE), gray relational analysis (GRA), multi-objective optimization on the basis of ratio analysis (MOORA), conflict analysis model (CAM), combinative distance based assessment (CODAS), multiplicative multi-objective ratio analysis (MULTIMOORA), best worst method (BWM), additive ratio assessment (ARAS), step-wise weight assessment ratio analysis (SWARA), complex proportional assessment (COPRAS), weighted additive sum product assessment (WASPAS), evaluation based on distance from average solution (EDAS), and MABAC. Simple additive weighting (SAW; also known as the weighted sum method) is also one of the most commonly used aggregating methods in the MCDM process. In addition, various types of operators can be used to obtain a composite index to compare the overall performance of the alternatives, and the ordered weighted averaging (OWA) operator is a commonly used operator.

2.3. Grouping

In the MCDM process, grouping or clustering is required when the main purpose of benchmarking is to compare the subjects [11]. Various grouping methods are available, such as PCA, cluster analysis (CA), multiple correspondence analysis (MCA), singular value decomposition (SVD), semi-discrete decomposition (SDD), fuzzy C-means (FCM), RSR, non-negative matrix factorization (NMF), and principal component analysis of scores of relative to an identified distribution integral transformation (PRIDIT). CA is the most popular method employed in a wide variety of scientific fields.

2.4. Research gaps

2.4.1. MCDM application

Previous research on the development of MCDM approaches has established a firm foundation for dealing with real-world decision-making problems in various fields. The efficiency, sensitivity, and uncertainty of final decisions resulting from the various options of different models remain a challenge for DMs. This is particularly true for small alternative samples. Moreover, previous approaches usually end with aggregating (i.e., ranking of the alternatives) without considering further step grouping, which is essential for more effective decision-making. Given these gaps, this study proposes a hybrid MCDM model with Monte Carlo simulation that integrates weighting, aggregating, and grouping with considerable stability, reliability, and applicability to address MCDM issues in the context of complex and compounded sensitivity and uncertainty.

2.4.2. Selecting MABAC

The MABAC approach was first proposed by Pamučar and Ćirović [18]. Since then, it has been extended in different ways (e.g., [19,20]). From our perspective, MABAC is a more effective method that can be used in MCDM cases, as demonstrated in a previous study [18]. In addition, it has three main advantages [21]: (1) it is propelled by a simple mathematical apparatus with stable results; (2) it can generate comprehensive and reliable results and considers the potential values of gains and losses; and (3) it is a flexible and adaptive aggregating method that can be adapted to other approaches. However, the reliability and effectiveness of MABAC need to be demonstrated through its application in practical cases. Additionally, MABAC has not been widely applied in the case of road safety development, whereas other methods, such as VIKOR and TOPSIS, are commonly used (e.g., [11,22,23]). Therefore, considering the reliability and effectiveness of MABAC and its limited application in the MCDM cases related to road safety development, it was selected as the main approach for our comparative assessment.

2.4.3. Challenges in k-means clustering

K-means is a widely used clustering method for segmentation and belongs to the categories of iterative and hill-climbing methods [24]. However, this method has some limitations. The performance of k-means is highly dependent on its starting points (initial centroids) and converges to the nearest local optima without searching the entire solution space [25], whereas randomly initialized prototypes often yield results that converge to local rather than global optima [26]. In other words, different initial centroid selections lead to inconsistent final grouping results, as demonstrated by the simulations in this study. During the programming process, a random seed is often required to fix the initial centroid position for subsequent verification. Nevertheless, fixing the initial position by setting a specific seed of random numbers often mixes with subjective epistemes from DMs. Several solutions have been proposed to address these limitations. For example, a centroid can be produced systematically, which is consistent with the distribution of data [27], a series (18) of novel centroid selection approaches in k-means clustering has been proposed to improve the recommendation process for recommender systems [28], and the time complexity and initial centroid selection issue of the original k-means clustering has been solved by a two-phase algorithm [29]. These resolutions concentrate on determining a more reasonable initial centroid, and currently, there is no general approach that does not require a specific initial centroid. In addition, most resolutions are only applicable

to large-sample cases, and treatments for small-sample cases are lacking. In small samples, the ambiguous grouping of marginal individuals can confuse the final decision-making process. This study resolves this problem by introducing a Monte Carlo simulation to the k-means algorithm.

3. Data

The effectiveness of the proposed hybrid MCDM model is tested and demonstrated through an application of transport safety accomplishments observed in the ASEAN region.

3.1. Safety performance indicators

Safety performance indicators should serve as supporting tools for government officials, DMs, and policymakers to measure the factors that contribute to accidents, identify conditions associated with increased safety risks, and detail the structure of traffic accident patterns [30]. However, selecting a set of appropriate indicators that can comprehensively monitor the status and achievements of transport safety is difficult. To a large extent, this depends on the availability, reliability, and quality of the data collected [31], which must be regularly updated and refined.

Based on five main dimensions (human-vehicle-road-environment-management), Al-Haji [32] proposed a comprehensive set of safety performance indicators for ASEAN, which have been acknowledged internationally. These indicators can be categorized into direct (output) and indirect (input) dimensions. The indicators of the direct dimension include derived measures, such as fatality rates, which are considered good measures for explaining national transport safety accomplishments [31]. Indicators of the indirect dimension are individual means that have been approved to be capable of describing the accomplishment or development of a specific theme of transport safety [10,33]. For more details on the links between performance indicators and safety outcomes, refer to review reports [30,32,34].

In this study, we updated and refined the original set of indicators by adding enforcement measures that were unavailable at that time. The updated framework of safety performance indicators is illustrated in Fig. 1.

3.2. Data collection

This study uses data from various sources (e.g., international databases, international organizations, and governmental publications)[3] for the 10 ASEAN members[4] and Timor-Leste (TL). Specifically, A_{11}, A_{21}, A_{31}, and C_{11} are calculated based on road traffic fatalities, registered vehicles, total vehicles, and the population reported by the World Health Organization (WHO). B_{11}, B_{12}, B_{13}, C_{41}, C_{42}, C_{43}, and C_{44} are obtained directly from the WHO. C_{31}, C_{32}, C_{33}, and C_{34} are obtained from the World Bank databases. C_{12} and C_{22} are compiled from related information on vehicles and roads provided by the WHO. C_{21} is collected from various sources, including ASEAN stats, the United Nations Economic and Social Commission for Asia and the Pacific (ESCAP), the library of the Central Intelligence Agency, and the International Road Federation.

4. Methodology

4.1. CRITIC

Numerous MCDM methods have been designed to evaluate the performance of alternatives based on several indicators, and weighting methods are applied to measure their relative importance (e.g., [35,36]). The criteria weights quantify their importance and can significantly affect the outcome of the decision-making process. This study uses an objective weight method, CRITIC.

CRITIC is a comprehensive method for objectively weighting indicators based on the comparative intensity and conflict between indicators [37]. Comparison intensity refers to the internal variation of a single indicator and is expressed as a standard deviation. An indicator with less variation should be assigned relatively less significance when making multi-criteria decisions, given its low ability to reflect the discrepancies among alternatives. The conflict between indicators is expressed by correlation coefficients. Multicollinearity variables often cause significant information overlap and should be assigned less weight because they mutually impair the contribution to the alternatives' performance evaluation [38].

4.2. MABAC

The MABAC approach was first proposed by Pamučar and Ćirović [18]. Within the framework of MABAC analysis, the performance

[3] Data sources include Global Status Report on Road Safety (2009, 2013, 2015, 2018) released by WHO (see https://www.who.int/publications/i/item), World Bank Indicators Database (see https://data.worldbank.org/indicator), ASEAN Statistics Division (see https://data.aseanstats.org/), ESCAP Statistical Database (see https://www.unescap.org/stat/data/statdb/DataExplorer.aspx), CIA (see https://www.cia.gov/library/publications/resources/the-world-factbook/), and IRF (see https://worldroadstatistics.org/).

[4] ASEAN comprises ten Southeast Asian states: Brunei (BN), Cambodia (KH), Indonesia (ID), Laos (LA), Malaysia (MY), Myanmar (MM), Philippines (PH), Singapore (SG), Thailand (TH), and Vietnam (VN).

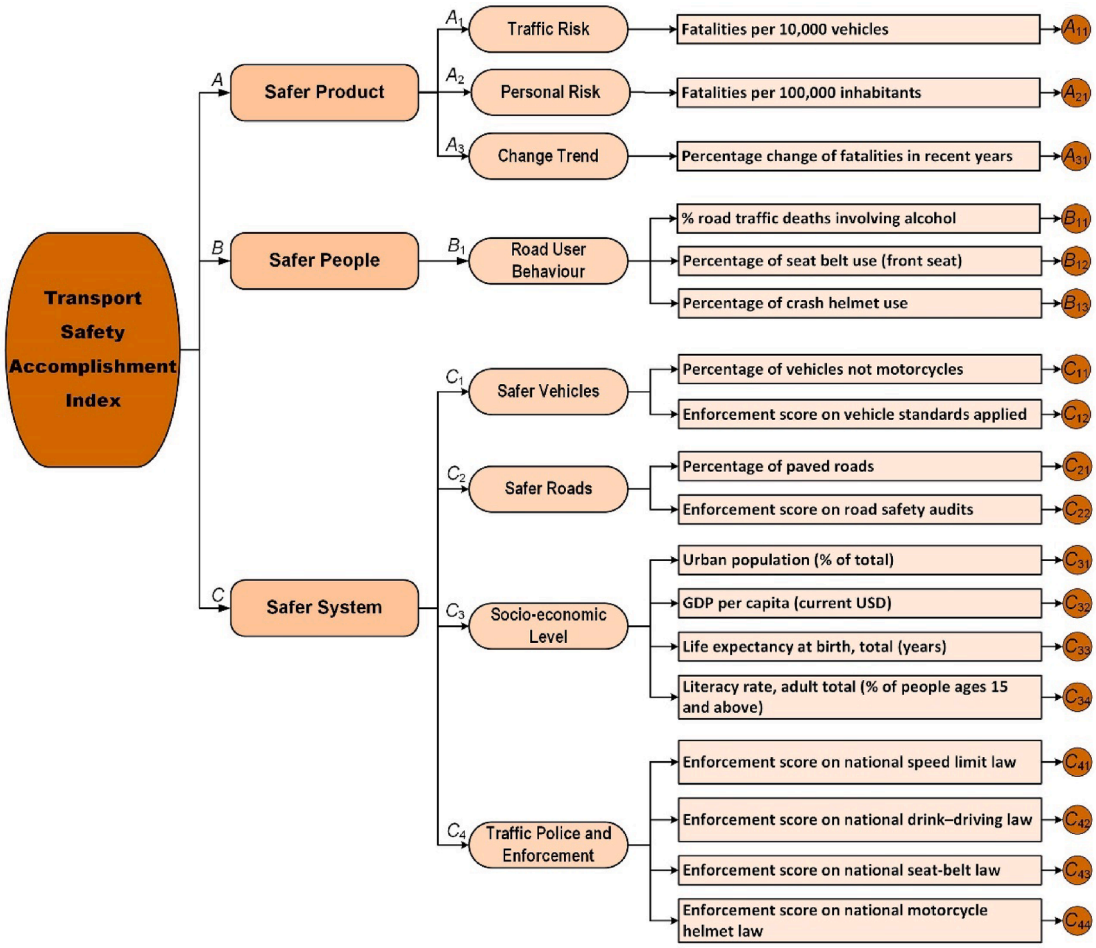


Fig. 1. Transport safety accomplishment index.

of every alternative is reflected by its position in an approximation region and its relative distance to the upper or lower boundaries [18]. This method provides an objective determination of the compromise ranking list and solutions with weight-stability intervals for preference stability. Alternatives in the region closer to the upper boundaries indicate better performance, whereas those closer to the lower boundaries indicate worse performance [19].

MABAC possesses ranking principles similar to other common aggregating methods (e.g. VIKOR and TOPSIS), where the relative distance of each alternative to positive and negative ideal solutions in linear space is used to establish the final ranking [5]. However, they differ not only in terms of the stability and efficiency of the final ranking but also in terms of the method of calculation. For example, TOPSIS computes the relative distance using the Euclidean distance, whereas MABAC quantifies the BAA with a multidimensional vector. Additionally, MABAC directly calculates the relative distance to the upper or lower boundary by measuring the deviation of each alternative from the BAA [39].

4.3. *K-means clustering*

K-means is a widely used clustering method for segmentation, introduced by the Polish mathematician Steinhaus [40] and developed by the American statistician MacQueen [24]. K-means is an unsupervised machine-learning method employing a top-down clustering approach. The dataset is divided into k groups based on similar characteristics, which is accomplished by minimizing the within-cluster variance of criterion [41]. The groups are created by minimizing the sum of square error (SSE) via an iterative algorithm, wherein the error is generally measured by the deviation of each alternative from the centroid, represented by the mean value vector of indicators for group members.

Although k-means is effective at minimizing the within-cluster variance, the selection of initial centroids can lead to inconsistent results, especially for peripheral points [26]. This inconsistency is demonstrated in the following data simulation where two types of data are generated: the first type of data is derived from 50 random observations of the standard normal distribution N (0,1), while the second type is derived from 50 random samples of the normal distribution N (1,1) (see Fig. 2). In our simulation, we divide the data

into two clusters. Different random seeds are set each time to generate distinct initial centroids. The results, presented in Fig. 2, illustrate that final categorizations are not always identical (the differences are circled).

The simulation suggests that the final cluster, to which some of the individuals belong, may be ambiguous owing to the difference in the initial center position. As most of the internal individuals in our simulation have consistent clustering results, and only five peripheral points exhibit ambiguity, k-means remain an effective method when serving big data cases where peripheral alternatives can be selectively neglected. However, as an unsupervised machine learning method, k-means may struggle to learn from insufficient information in small-sample cases. Big sample data are less likely to be affected by the variation in initial centroid selection, as only a few peripheral alternatives have clustering inconsistency, and most internal individuals are objectively allocated. However, small-sample data show a significant bias, especially when erratic alternatives account for a large proportion of the total sample. Our sample size for ASEAN road safety is not sufficiently large, with only 11 countries as alternatives to be analyzed, and the ambiguity of peripheral individuals' clustering strongly affects the results. We find several possible clustering results for the 11 countries, and this inconsistency complicates the decision-making process for determining the outcome.

To resolve this problem, we conduct a Monte Carlo simulation in which the k-means clustering process is executed 1,000 times without any special random seed settings for the initial centroid. Different types of grouping results are generated by calculating the frequency of each result. The result with the highest frequency is regarded as the final result. In addition, we calculate the probabilities of the grouping result with the highest frequency at the 1%, 5%, and 10% significance levels based on the normal distribution and compare them with the grouping result that has the second-highest frequency. This allows us to determine whether the final result is dominant over the second-most probable result at these significance levels.

4.4. CRITIC–MABAC–Kmeans with Monte Carlo simulation

We develop a new synthetic model for weighting, aggregating, and grouping, which combines three mathematical methodologies. The model includes a CRITIC weighting method that considers both the comparison intensity and information conflict to objectively obtain the aggravated ranking result using MABAC. As the alternatives evaluated by MABAC are points in an approximation region, k-means clustering is used to group these points based on their degree of aggregation in the region. Subsequently, we conduct a Monte Carlo simulation to determine the final results. The methodological flowchart is presented in Fig. 3.

The detailed steps are as follows:

Step 1: Decision matrix construction.

Our original dataset contains n major indicators to evaluate the m alternatives (the 11 countries in Southeast Asia). The value for the alternative i ($1 \leq i \leq m$) concerning indicator j ($1 \leq j \leq n$) is denoted by a_{ij}. This is illustrated by matrix A.

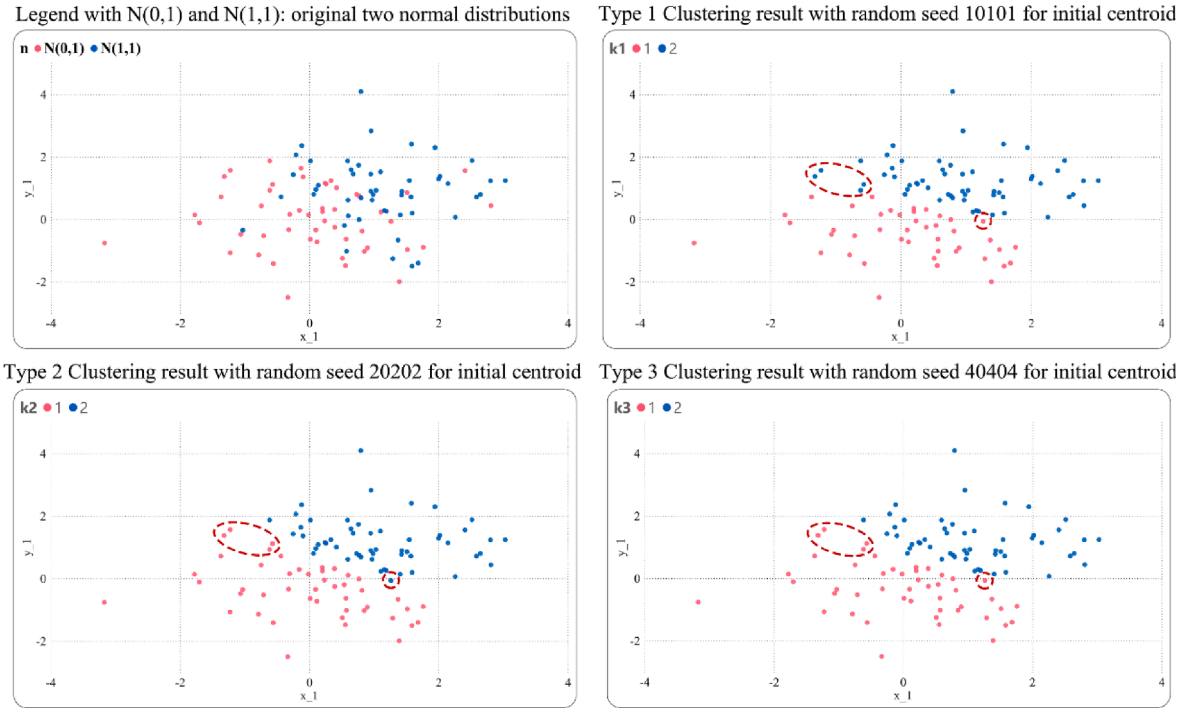


Fig. 2. Simulation process of k-means clustering with different initial centroids.

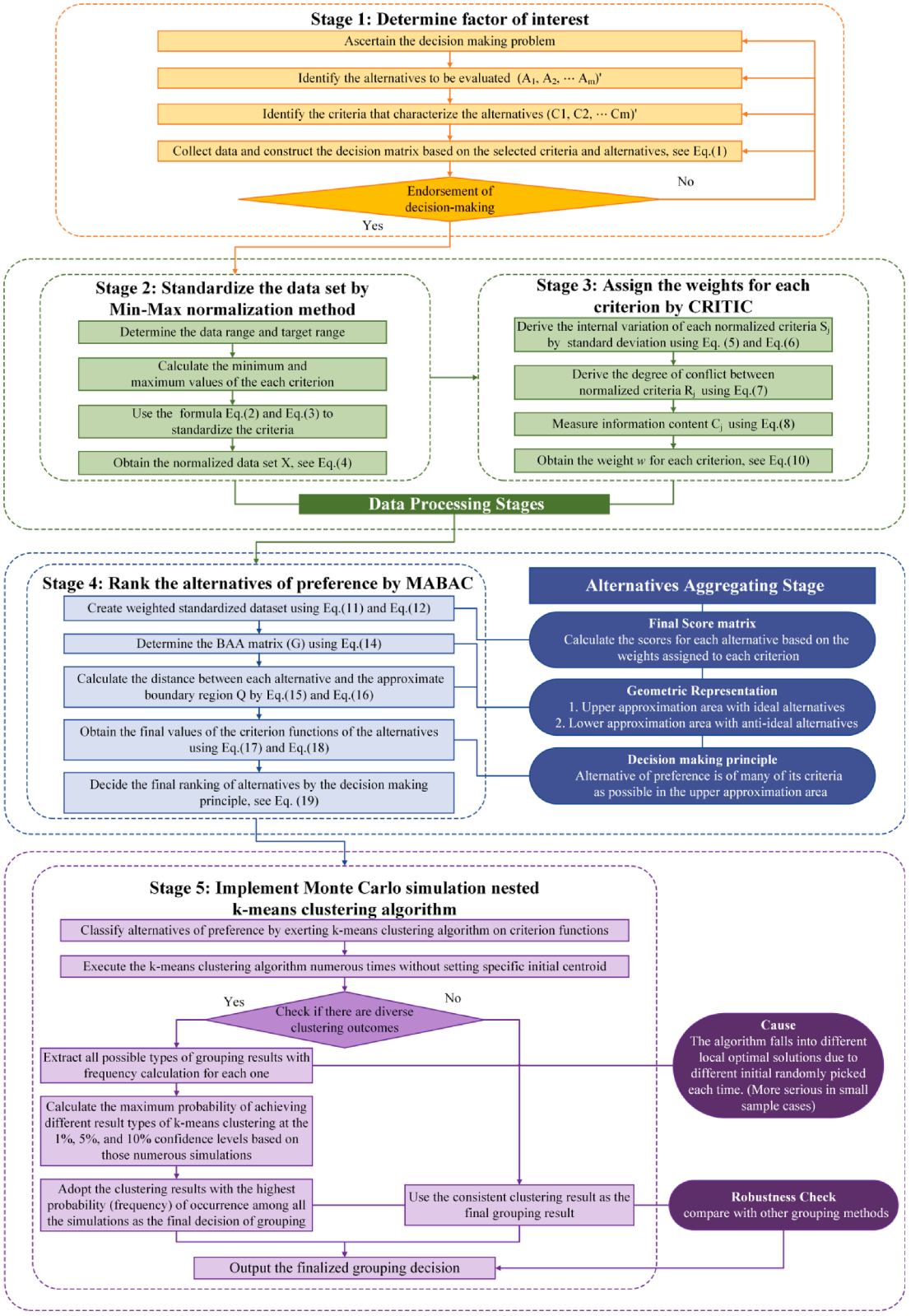


Fig. 3. Methodological flowchart of the proposed hybrid MCDM model.

$$A = \begin{array}{c} \\ A_1 \\ \cdots \\ A_m \end{array} \begin{array}{c} \begin{array}{ccc} C_1 & \cdots & C_n \end{array} \\ \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \end{array} \tag{1}$$

Step 2: Data normalization.

As the indicators in the dataset have different magnitudes, a data standardization process is applied to transform various attribute dimensions of the indicators into non-dimensional attributes to mix them. As the indicators acquire the same magnitude, a comparison across criteria is feasible [42].

Our ASEAN study includes 18 indicators to describe the 11 countries, such that $m = 18$ and $n = 11$. For the negative indicators (C_1, C_2, C_3, and C_4 in this case study), the standardized values are based on the following formula:

$$X_{ij} = \frac{\max_i(a_{ij}) - a_{ij}}{\max_i(a_{ij}) - \min_i(a_{ij})} \tag{2}$$

This process not only plays a role in data normalization but also helps convert these negative indicators into a positive forward direction [43].

For the positive indicators (C_5, C_6, C_7, ⋯, C_{18}), the standardized values are based on the following formula:

$$X_{ij} = \frac{a_{ij} - \min_i(a_{ij})}{\max_i(a_{ij}) - \min_i(a_{ij})} \tag{3}$$

Based on the illustrated calculation, the original dataset A can be converted into a normalized dataset X, in which all elements acquire the same attitude between 0 and 1 and become positive forward.

$$X = \begin{array}{c} \\ A_1 \\ \cdots \\ A_m \end{array} \begin{array}{c} \begin{array}{ccc} C_1 & \cdots & C_n \end{array} \\ \begin{pmatrix} X_{11} & \cdots & X_{1n} \\ \vdots & \ddots & \vdots \\ X_{m1} & \cdots & X_{mn} \end{pmatrix} \end{array} \tag{4}$$

Step 3: Assign the weights for each indicator.

The weight calculation is established upon the normalized data set X, within which the standard deviation for each criterion (C_1, C_2, ⋯, C_{18}) is accessed.

$$\bar{x}_j = \frac{1}{m} \sum_{i=1, x \in x_j}^{m} x_i \tag{5}$$

$$S_j = \sqrt{\sum_{i=1, x \in x_j}^{m} (x_i - \bar{x}_j)^2} \tag{6}$$

S_j denotes the standard deviation of indicator j. The conflict between the indicators is reflected in the correlation coefficients.

$$R_j = \sum (1 - |r_{jm}|) \tag{7}$$

where r_{jm} denotes the correlation coefficients between indicators j and m. For all indicators, the correlation coefficients are calculated between every two pairs. The value of R_j is calculated as the sum of 1 minus each correlation coefficient. This formula suggests that indicators that are highly correlated with each other have overlapping information in explaining the socioeconomic phenomenon and should be assigned less weight (importance). In our study, the absolute value of the calculated correlation coefficient r_{jm} should be used because a negative correlation indicates information overlap.

As the weight reflects the relative importance of each indicator, we believe that indicators with more information content can more significantly reflect the objects being explained. The formula for measuring information content is as follows:

$$C_j = S_j \sum (1 - |r_{jm}|) = S_j \cdot R_j \tag{8}$$

Based on the information content, the objective weight is calculated as:

$$w_j = \frac{C_j}{\sum_{j=1}^{m} C_j} \tag{9}$$

Therefore, a weight vector for each indicator is created, where w_j denotes the weight for the indicator j ($1 \le j \le n$):

$$w = (w_1 \quad w_2 \quad \cdots \quad w_n) \tag{10}$$

Step 4: Weighted standardized dataset creation.

The elements of the weighted standardized matrix are calculated based on the following expression:

$$h_{ij} = w_j \cdot (1 + X_{ij}) \tag{11}$$

where X_{ij} is the element from the normalized dataset X and w_j is the objective weight for indicator j obtained by CRITIC. Therefore, the weighted standardized matrix H is obtained.

$$H = \begin{pmatrix} w_1 \cdot (1 + X_{11}) & w_2 \cdot (1 + X_{12}) & \dots & w_m \cdot (1 + X_{1m}) \\ w_1 \cdot (1 + X_{21}) & w_2 \cdot (1 + X_{22}) & \dots & w_m \cdot (1 + X_{2m}) \\ \vdots & \vdots & \ddots & \vdots \\ w_1 \cdot (1 + X_{n1}) & w_2 \cdot (1 + X_{n2}) & \cdots & w_m \cdot (1 + X_{nm}) \end{pmatrix} = \begin{pmatrix} h_{11} & h_{12} & \cdots & h_{1n} \\ h_{21} & h_{22} & \cdots & h_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ h_{m1} & h_{m2} & \cdots & h_{mn} \end{pmatrix} \tag{12}$$

Step 5: Determining the BAA matrix (G).

The MABAC method focuses on the relative position of each alternative in the approximation region, and the quantified BAA for each indicator is determined as follows:

$$G = \begin{pmatrix} g_1 & g_2 & \cdots & g_n \end{pmatrix} \tag{13}$$

The element of indicator j in BAA matrix G is calculated using the following equation, as we have m alternatives:

$$g_j = \left(\prod_{i=1}^{m} h_{ij}^{1/m} \right) \tag{14}$$

where h_{ij} denotes the elements of the weighted standardized matrix H.

Step 6: Calculating the distance between each alternative and the approximate boundary region.

As the BAA is successfully quantified by the matrix G, the distance of the alternatives to the BAA (q_{ij}) is measured by the difference between the elements in the weighted standardized matrix H and the BAA matrix (G).

$$q_{ij} = h_{ij} - g_j \tag{15}$$

$$Q = \begin{pmatrix} q_{11} & q_{12} & \cdots & q_{1n} \\ q_{21} & q_{22} & \cdots & q_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ q_{m1} & q_{m2} & \cdots & q_{mn} \end{pmatrix} = \begin{pmatrix} h_{11} & h_{12} & \cdots & h_{1n} \\ h_{21} & h_{22} & \cdots & h_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ h_{m1} & h_{m2} & \cdots & h_{mn} \end{pmatrix} - \begin{pmatrix} g_1 & g_2 & \cdots & g_m \\ g_1 & g_2 & \cdots & g_m \\ \cdots & \cdots & \cdots & \cdots \\ g_1 & g_2 & \cdots & g_m \end{pmatrix} \tag{16}$$

By calculating the sum of the elements of matrix Q by rows, we obtain the final values of the criterion functions of the alternatives.

$$S_i = \sum_{j=1}^{n} q_{ij} \tag{17}$$

$$S = \begin{pmatrix} q_{11} + q_{12} + \cdots + q_{1n} \\ q_{21} + q_{22} + \cdots + q_{2n} \\ \cdots\cdots \\ q_{m1} + q_{m2} + \cdots + q_{mn} \end{pmatrix} = \begin{pmatrix} S_1 \\ S_2 \\ \cdots \\ S_m \end{pmatrix} \tag{18}$$

The elements in the column vector S are the row sums of matrix Q. For all alternatives, the one with a higher S value indicates better performance.

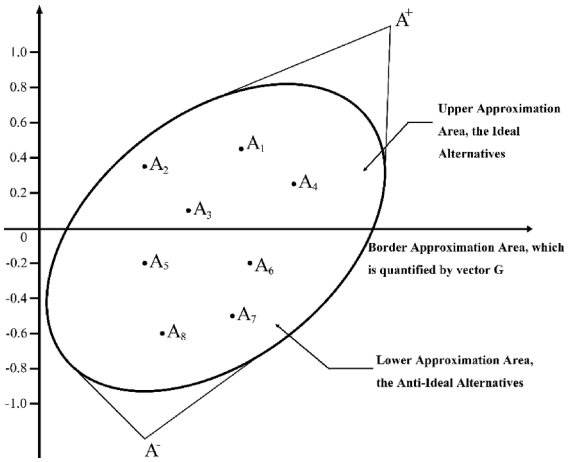


Fig. 4. Geometric representation of MABAC.

Table 1
Comparison of rankings across different aggregating and weighting methods under **MinMax** normalization (**2018**).

Country	**2018**											
	MinMax-CRITIC-MABAC	MinMax-Entropy-MABAC	Deviation	MinMax-CRITIC-TOPSIS	MinMax-Entropy-TOPSIS	Deviation	MinMax-CRITIC-RSR	MinMax-Entropy-RSR	Deviation	MinMax-CRITIC-VIKOR	MinMax-Entropy-VIKOR	Deviation
BN	2	2	0	2	2	0	2	2	0	2	2	0
ID	3	4	1	3	4	1	3	4	1	3	4	1
KH	9	9	0	9	9	0	9	9	0	9	9	0
LA	11	11	0	11	11	0	11	11	0	11	11	0
MM	8	8	0	8	8	0	8	8	0	8	8	0
MY	4	3	1	5	1	4	4	3	1	4	3	1
PH	6	7	1	6	7	1	6	7	1	6	7	1
SG	1	1	0	1	3	2	1	1	0	1	1	0
TH	7	6	1	7	5	2	7	6	1	7	6	1
TL	10	10	0	10	10	0	10	10	0	10	10	0
VN	5	5	0	4	6	2	5	5	0	5	5	0
Sum	—	—	4	—	—	12	—	—	4	—	—	4
r_s	0.98		—	0.86		—	0.98		—	0.98		—
r_w	0.98		—	0.81		—	0.98		—	0.98		—
WS	0.97		—	0.84		—	0.97		—	0.97		—

Note: r_s = Spearman coefficient; r_w = weighted rank measure of correlation; *WS* = rank similarity coefficient.

Table 2
Comparison of rankings across different aggregating and weighting methods under **MinMax** normalization (**2015**).

Country	**2015**											
	MinMax-CRITIC-MABAC	MinMax-Entropy-MABAC	Deviation	MinMax-CRITIC-TOPSIS	MinMax-Entropy-TOPSIS	Deviation	MinMax-CRITIC-RSR	MinMax-Entropy-RSR	Deviation	MinMax-CRITIC-VIKOR	MinMax-Entropy-VIKOR	Deviation
BN	1	1	0	1	1	0	1	1	0	1	1	0
ID	4	4	0	4	5	1	4	4	0	4	4	0
KH	11	10	1	11	11	0	11	10	1	11	10	1
LA	8	8	0	8	8	0	8	8	0	8	8	0
MM	10	11	1	9	10	1	10	11	1	10	11	1
MY	3	3	0	3	2	1	3	3	0	3	3	0
PH	5	6	1	6	6	0	5	6	1	5	6	1
SG	2	2	0	2	3	1	2	2	0	2	2	0
TH	7	7	0	7	7	0	7	7	0	7	7	0
TL	9	9	0	10	9	1	9	9	0	9	9	0
VN	6	5	1	5	4	1	6	5	1	6	5	1
Sum	—	—	4	—	—	6	—	—	4	—	—	4
r_s	0.98		—	0.97		—	0.98		—	0.98		—
r_w	0.99		—	0.97		—	0.99		—	0.99		—
WS	0.99		—	0.94		—	0.99		—	0.99		—

Note: r_s = Spearman coefficient; r_w = weighted rank measure of correlation; *WS* = rank similarity coefficient.

Table 3
Comparison of rankings across different aggregating and weighting methods under **MinMax** normalization (**2013**).

Country	**2013**											
	MinMax-CRITIC-MABAC	MinMax-Entropy-MABAC	Deviation	MinMax-CRITIC-TOPSIS	MinMax-Entropy-TOPSIS	Deviation	MinMax-CRITIC-RSR	MinMax-Entropy-RSR	Deviation	MinMax-CRITIC-VIKOR	MinMax-Entropy-VIKOR	Deviation
BN	1	2	1	1	2	1	1	2	1	1	2	1
ID	4	5	1	4	5	1	4	5	1	4	5	1
KH	8	9	1	7	9	2	8	9	1	8	9	1
LA	9	8	1	8	8	0	9	8	1	9	8	1
MM	10	11	1	10	11	1	10	11	1	10	11	1
MY	5	4	1	5	4	1	5	4	1	5	4	1
PH	6	6	0	6	6	0	6	6	0	6	6	0
SG	2	1	1	2	1	1	2	1	1	2	1	1
TH	7	7	0	9	7	2	7	7	0	7	7	0
TL	11	10	1	11	10	1	11	10	1	11	10	1
VN	3	3	0	3	3	0	3	3	0	3	3	0
Sum	—	—	8	—	—	10	—	—	8	—	—	8
r_s	0.96		—	0.94		—	0.96		—	0.96		—
r_w	0.97		—	0.95		—	0.97		—	0.97		—
WS	0.91		—	0.90		—	0.91		—	0.91		—

Note: r_s = Spearman coefficient; r_w = weighted rank measure of correlation; *WS* = rank similarity coefficient.

Table 4
Comparison of rankings across different aggregating and weighting methods under **MinMax** normalization (**2009**).

Country	**2009**											
	MinMax-CRITIC-MABAC	MinMax-Entropy-MABAC	Deviation	MinMax-CRITIC-TOPSIS	MinMax-Entropy-TOPSIS	Deviation	MinMax-CRITIC-RSR	MinMax-Entropy-RSR	Deviation	MinMax-CRITIC-VIKOR	MinMax-Entropy-VIKOR	Deviation
BN	3	3	0	6	4	2	3	3	0	3	3	0
ID	6	5	1	7	6	1	6	5	1	6	5	1
KH	11	11	0	11	11	0	11	11	0	11	11	0
LA	9	9	0	8	8	0	9	9	0	9	9	0
MM	7	6	1	4	5	1	7	6	1	7	6	1
MY	2	2	0	2	2	0	2	2	0	2	2	0
PH	5	7	2	5	7	2	5	7	2	5	7	2
SG	1	1	0	1	1	0	1	1	0	1	1	0
TH	8	8	0	9	9	0	8	8	0	8	8	0
TL	10	10	0	10	10	0	10	10	0	10	10	0
VN	4	4	0	3	3	0	4	4	0	4	4	0
Sum	—	—	4	—	—	6	—	—	4	—	—	4
r_s	0.97		—	0.95		—	0.97		—	0.97		—
r_w	0.97		—	0.95		—	0.97		—	0.97		—
WS	0.99		—	0.97		—	0.99		—	0.99		—

Note: r_s = Spearman coefficient; r_w = weighted rank measure of correlation; *WS* = rank similarity coefficient.

This provides a geometric explanation of MABAC [18]. Alternative A_i could belong to the BAA (G), upper approximation area (G^+), or lower approximation area (G^-), that is, $A_i \in \{G \vee G^+ \vee G^-\}$. The upper approximation area (G^+) is the area that contains the ideal alternative (A^+), while the lower approximation area (G^-) is the area that contains the anti-ideal alternative (A^-).

$$A_i \in \begin{Bmatrix} G^+ if\, q_{ij} > 0 \\ G if\, q_{ij} = 0 \\ G^- if\, q_{ij} < 0 \end{Bmatrix} \tag{19}$$

Fig. 4 divides the total approximation region into two parts, separated by a horizontal BAA. A_1 to A_4 are regarded as ideal alternatives as they are in the upper approximation area, whereas A_5 to A_8 are anti-ideal alternatives in the lower approximation area. An alternative's performance is determined by its proximity to the upper or lower boundary. The relative distance of alternative i with the respective indicator j is q_{ij}, with the sign indicating the area to which the alternative belongs and the absolute value indicating its distance from the BAA (i.e., how close it is to the upper and lower boundaries).

The evaluation of alternative A_i considers several indicators, and to be the best in the set, it needs to have as many of its indicators as possible in the upper approximation area (G^+). Therefore, a comprehensive analysis requires summing up the relative distances (q_{ij}) of all indicators for each alternative.

Step 7: K-means and Monte Carlo simulation.

Based on the degree of aggregation of the univariate points in the approximation region, the alternatives are sorted into k groups (clusters), where k is predefined. The k-means clustering process is expressed as follows:

The alternatives in the dataset can be partitioned into K disjoint sets of $\{M_1, \cdots, M_K\}$, where $M_m \cap M_h = \varnothing (\forall m \neq h)$ and $M_1 \cup M_2 \cup \cdots \cup M_K = \{1, \cdots, n\}$. Essentially, each alternative i has only one cluster to which it belongs, and $i \in M_h$ implies that the similarity of characteristics (indicators) brings together alternatives in group h. The centroid of each group is determined using the mean values of the indicators in linear space. For example, the centroid of cluster h is defined as follows:

$$m_h = \frac{1}{n_h} \sum_{i \in M_h} x_i \tag{20}$$

where n_h denotes the number of alternatives in group h and x_i specifies the indicators of alternatives in cluster h. Clearly, the centroid m_h is explained by the indicator mean values of the group members. For any alternative i in group k, the error term is measured by its deviation from the centroid in the linear space $(x_i - m_h)$.

When the total value within the SSE for all k clusters is minimized, alternatives with similar characteristics converge.

$$\min_{M_1 \cdots M_K} SSE = \min \sum_{h=1}^{K} \sum_{i \in M_h} \|x_i - m_h\| \tag{21}$$

An iterative algorithm helps find optimal solutions for minimizing SSE. The process involves the following steps:

1. Randomly select the centroid of the cluster.
2. Repeat the following two steps until the convergence is achieved:

a) Reassign each alternative to its nearest cluster.

b) Update the new centroid based on the reallocation.

The SSE of the full sample must decrease with each iteration of the loop because all observations are assigned to a closer cluster at each iteration. When the gradient descent value of the SSE is extremely small, a local minimum solution is achieved [24]. In our study, the number of clusters k is predetermined to be three, and the unique characteristic is the final S vector acquired at the end of MABAC. The absolute distance is used to measure the degree of aggregation.

However, an iterative algorithm can only determine the SSE local minimum, which is not necessarily a global minimum solution. A disparate local minimum solution is achieved if the initial centroid is selected differently, leading to divergence in the final result [25]. In practice, we perform the k-means algorithm program 1000 times without specifically setting the initial centroid and extract all possible types of grouping results with frequency calculation for each one. Combined with probability calculations at 1%, 5%, and 10% significance levels, the final grouping results are thus determined.

5. Results and discussion

5.1. Stability of ranking

Internal stability refers to the extent to which the results of a specific aggregating method are disrupted by choosing different normalization and weighting methods. External stability refers to the consistency of the results when compared with other aggregating methods.

5.1.1. Internal stability

Table 1, Table 2, Table 3, and Table 4 present the corresponding rankings obtained by different aggregating methods (MABAC, TOPSIS, RSR, and VIKOR) across weighting methods (CRITIC and Entropy) under **MinMax** normalization over four years (2018, 2015,

Table 5
Comparison of rankings across different aggregating and weighting methods under **Vector** normalization (**2018**).

Country	**2018**											
	Vector-CRITIC-MABAC	Vector-Entropy-MABAC	Deviation	Vector-CRITIC-TOPSIS	Vector-Entropy-TOPSIS	Deviation	Vector-CRITIC-RSR	Vector-Entropy-RSR	Deviation	Vector-CRITIC-VIKOR	Vector-Entropy-VIKOR	Deviation
BN	3	2	1	4	3	1	3	2	1	3	2	1
ID	6	6	0	6	6	0	5	6	1	5	6	1
KH	10	11	1	10	11	1	9	9	0	9	9	0
LA	11	10	1	11	9	2	11	11	0	11	11	0
MM	8	8	0	9	10	1	8	8	0	8	8	0
MY	1	1	0	1	1	0	2	1	1	2	1	1
PH	4	5	1	3	5	2	4	4	0	4	4	0
SG	2	3	1	2	2	0	1	3	2	1	3	2
TH	5	4	1	5	4	1	7	5	2	6	5	1
TL	9	9	0	8	8	0	10	10	0	10	10	0
VN	7	7	0	7	7	0	6	7	1	7	7	0
Sum	—	—	6	—	—	8	—	—	8	—	—	6
r_s	0.97		—	0.95		—	0.95		—	0.96		—
r_w	0.97		—	0.95		—	0.93		—	0.94		—
WS	0.94		—	0.95		—	0.85		—	0.85		—

Note: r_s = Spearman coefficient; r_w = weighted rank measure of correlation; *WS* = rank similarity coefficient.

Table 6
Comparison of rankings across different aggregating and weighting methods under **Vector** normalization (**2015**).

Country	**2015**											
	Vector-CRITIC-MABAC	Vector-Entropy-MABAC	Deviation	Vector-CRITIC-TOPSIS	Vector-Entropy-TOPSIS	Deviation	Vector-CRITIC-RSR	Vector-Entropy-RSR	Deviation	Vector-CRITIC-VIKOR	Vector-Entropy-VIKOR	Deviation
BN	2	1	1	2	1	1	1	1	0	1	1	0
ID	5	6	1	5	6	1	5	5	0	5	5	0
KH	10	10	0	9	9	0	10	10	0	10	10	0
LA	9	9	0	11	11	0	9	9	0	9	9	0
MM	11	11	0	10	10	0	11	11	0	11	11	0
MY	4	3	1	4	2	2	4	2	2	4	2	2
PH	1	2	1	1	3	2	3	3	0	3	3	0
SG	3	4	1	3	4	1	2	4	2	2	4	2
TH	6	5	1	6	5	1	7	6	1	7	6	1
TL	8	8	0	8	8	0	8	8	0	8	8	0
VN	7	7	0	7	7	0	6	7	1	6	7	1
Sum	—	—	6	—	—	8	—	—	6	—	—	6
r_s	0.97		—	0.95		—	0.95		—	0.95		—
r_w	0.96		—	0.92		—	0.94		—	0.94		—
WS	0.89		—	0.83		—	0.92		—	0.92		—

Note: r_s = Spearman coefficient; r_w = weighted rank measure of correlation; *WS* = rank similarity coefficient.

Table 7
Comparison of rankings across different aggregating and weighting methods under **Vector** normalization (**2013**).

Country	**2013**											
	Vector-CRITIC-MABAC	Vector-Entropy-MABAC	Deviation	Vector-CRITIC-TOPSIS	Vector-Entropy-TOPSIS	Deviation	Vector-CRITIC-RSR	Vector-Entropy-RSR	Deviation	Vector-CRITIC-VIKOR	Vector-Entropy-VIKOR	Deviation
BN	3	2	1	3	3	0	1	2	1	1	2	1
ID	7	6	1	8	6	2	6	6	0	6	6	0
KH	10	10	0	9	9	0	9	10	1	9	10	1
LA	9	9	0	10	11	1	8	9	1	8	9	1
MM	11	11	0	11	10	1	11	11	0	11	11	0
MY	4	4	0	4	4	0	4	4	0	4	4	0
PH	2	1	1	1	1	0	3	3	0	3	3	0
SG	1	3	2	2	2	0	2	1	1	2	1	1
TH	5	5	0	6	5	1	7	5	2	7	5	2
TL	8	8	0	5	7	2	10	8	2	10	8	2
VN	6	7	1	7	8	1	5	7	2	5	7	2
Sum	—	—	6	—	—	8	—	—	10	—	—	10
r_s	0.96		—	0.95		—	0.93		—	0.93		—
r_w	0.95		—	0.96		—	0.93		—	0.93		—
WS	0.85		—	0.98		—	0.92		—	0.92		—

Note: r_s = Spearman coefficient; r_w = weighted rank measure of correlation; *WS* = rank similarity coefficient.

2012, and 2009), along with the ranking similarity, for comparison. Table 5, Table 6, Table 7, and Table 8 present the corresponding rankings under **Vector** normalization over four years (2018, 2015, 2012, and 2009), along with the ranking similarity.

The results presented in Table 1, Table 2, Table 3, and Table 4 demonstrate two advantages of MABAC as an effective aggregating method for MCDM cases. First, MABAC is stable in terms of the sensitivity of the result to the weights of the criteria, as shown by the smaller deviations compared with TOPSIS for different weighting methods (MinMax-CRITIC vs. MinMax-entropy). The fact that the sum of deviation in MABAC is significantly smaller than its counterparts in TOPSIS in the four years preliminarily indicates that MABAC is less inclined to be impacted by different weight methods than TOPSIS. The comparisons are extended to two other aggregating methods, RSR and VIKOR. As the sum of the deviation in MABAC after changing the weight methods is smaller than that of RSR and VIKOR, it can be preliminarily deduced that MABAC is at least as effective and stable as RSR and VIKOR, given its minor sensitivity to different weights. More compelling comparisons involve the use of typical coefficients of ranking similarities in decision-making problems. The similarity of the final rankings by MABAC under different weight methods (CRITIC vs. Entropy) is higher than that of TOPSIS and is equal to RSR and VIKOR because all three coefficients (r_s, r_w, and WS)[5] for MABAC are higher than those for TOPSIS and equal to those for RSR and VIKOR. The coefficients indicate that MABAC is more stable than TOPSIS, RSR, or VIKOR.

To obtain more robust results, the contrast is extended to other data normalization methods (Vector programming), as shown in Table 5, Table 6, Table 7, and Table 8. In this case, MABAC has less deviation than the other aggregating methods: TOPSIS, RSR, and VIKOR. Ubiquitously, MABAC reports a higher similarity under different weight methods (Vector-CRITIC vs. Vector-Entropy) than the other three aggregating methods because of its higher similarity coefficients. Overall, it can be concluded that MABAC is a more stable aggregating method that is less susceptible to weight changes.

5.1.2. *External stability*

MABAC exhibits consistency, as demonstrated by the small number of inconsistent ranks compared to the other three aggregating methods. Table 9 further supports the consistency of MABAC by comparing it with the other aggregating methods (TOPSIS, RSR, and VIKOR) for a given weight method (MinMax-CRITIC). These results show that MABAC is highly consistent with RSR and VIKOR, whereas TOPSIS loses consistency with some alternatives. In addition, Table 9 reports the similarity coefficients between MABAC and each of the other three aggregating methods, suggesting that MABAC ranking is highly similar to TOPSIS, RSR, and VIKOR because the absolute values of these coefficients are all close to 1. This high degree of consistency holds even after the weighting method is changed. Table 10 reports another comparison between MABAC ranking and those from the other aggregating methods (TOPSIS, RSR, and VIKOR) under the MinMax-Entropy. In this case, MABAC rankings still show a high degree of similarity with the rankings of the other methods in terms of high-similarity coefficients.

Overall, the MABAC method is a reliable tool for MCDM cases, effectively reducing the ambiguity caused by different weighting methods, and is consistent with other common aggregating methods. Such extrapolation is congruous with the findings of Pamučar and Ćirović [18], which demonstrate the effectiveness of MABAC through sensitivity and consistency analyses. The current study shows that MABAC is less likely to be disrupted than TOPSIS, RSR, and VIKOR when the weights of the indicators are changed. Furthermore, MABAC exhibited stability in the results depending on changes in the unit of measurement used to express the value of individual indicators and the formulation of the indicator in cases where the same indicator can be expressed by a positive forward or negative forward. In contrast, TOPSIS, RSR, and VIKOR failed in these cases.

5.2. *Reliability of grouping*

MCDM cases on road safety development often require a grouping process involving categorization based on the ranking results, which can provide a "best-in-class" benchmark for other alternatives in the same group (e.g., [11,44,45]). Therefore, our study uses a k-means algorithm dependent on the criteria function (S vector) for country categorization and a Monte Carlo simulation to resolve inconsistencies in the results. Table 11 shows the results of k-means clustering for countries in 2018, coupled with the results of PCA grouping and hierarchical clustering analysis (HCA) with centroid linkages, which have been widely applied in previous studies. Table 12 presents the results for the k-means algorithm executed 1000 times without setting the initial centroid.

The Monte Carlo simulation and k-means clustering suggest four types of results, in which k4s_mean has the highest frequency of occurrences and is selected as the final grouping result. Table 12 presents the maximum probability of achieving different result types at the 1%, 5%, and 10% confidence levels based on 1,000 simulations. These probabilities are calculated using the normal distribution and can be regarded as a reverse hypothesis test. When the sample size, significance level, and proportion of each result type are known, we can compute the maximum probability at which a hypothesis test can be performed without rejection at these significance levels. For instance, based on our simulation, we can be 99% confident that the probability of k4s_mean in the population is 56%. To verify the accuracy of the Monte Carlo simulation, we provide the scatter of the criteria function (S vector) for the 11 countries, as the k-means clustering is based on the aggregation degree for indicators in the linear space. The k4s_mean classification is well reflected in Fig. 5 (for 2018) in which "BN" and "SG" seem to cluster together and can be included in the "Excellent Group," while "LA," "TL," "KH," and "MM" are treated as members of the "Poor Group." The remaining countries are assembled at the medium place and allocated to the "Intermediate Group."

In addition, we use PCA grouping as a comparison method in the robustness test for our k-means results. Table 11 shows that the

[5] The procedure for calculating the ranking similarity can be found in Appendix A.

Table 8
Comparison of rankings across different aggregating and weighting methods under **Vector** normalization (**2009**).

Country	**2009**											
	Vector-CRITIC-MABAC	Vector-Entropy-MABAC	Deviation	Vector-CRITIC-TOPSIS	Vector-Entropy-TOPSIS	Deviation	Vector-CRITIC-RSR	Vector-Entropy-RSR	Deviation	Vector-CRITIC-VIKOR	Vector-Entropy-VIKOR	Deviation
BN	4	4	0	5	5	0	4	4	0	4	4	0
ID	6	6	0	7	7	0	6	6	0	6	6	0
KH	11	11	0	11	11	0	11	11	0	11	11	0
LA	9	9	0	9	9	0	9	9	0	9	9	0
MM	7	7	0	6	6	0	7	7	0	7	7	0
MY	3	3	0	3	2	1	2	2	0	2	2	0
PH	2	2	0	1	3	2	3	3	0	3	3	0
SG	1	1	0	2	1	1	1	1	0	1	1	0
TH	8	8	0	8	8	0	8	8	0	8	8	0
TL	10	10	0	10	10	0	10	10	0	10	10	0
VN	5	5	0	4	4	0	5	5	0	5	5	0
Sum	—	—	0	—	—	4	—	—	0	—	—	0
r_s	1.00		—	0.97		—	1.00		—	1.00		—
r_w	1.00		—	0.95		—	1.00		—	1.00		—
WS	1.00		—	0.86		—	1.00		—	1.00		—

Note: r_s = Spearman coefficient; r_w = weighted rank measure of correlation; *WS* = rank similarity coefficient.

Table 9
Comparison of rankings by different aggregating methods in four years under **MinMax-CRITIC**.

Country	2018				2015				2013				2009			
	MABAC	TOPSIS	RSR	VIKOR	MABAC	TOPSIS	RSR	VIKOR	MABAC	TOPSIS	RSR	VIKOR	MABAC	TOPSIS	RSR	VIKOR
BN	2	2	2	2	1	1	1	1	1	1	1	1	3	6	3	3
ID	3	3	3	3	4	4	4	4	4	4	4	4	6	7	6	6
KH	9	9	9	9	11	11	11	11	8	7	8	8	11	11	11	11
LA	11	11	11	11	8	8	8	8	9	8	9	9	9	8	9	9
MM	8	8	8	8	10	9	10	10	10	10	10	10	7	4	7	7
MY	4	5	4	4	3	3	3	3	5	5	5	5	2	2	2	2
PH	6	6	6	6	5	6	5	5	6	6	6	6	5	5	5	5
SG	1	1	1	1	2	2	2	2	2	2	2	2	1	1	1	1
TH	7	7	7	7	7	7	7	7	7	9	7	7	8	9	8	8
TL	10	10	10	10	9	10	9	9	11	11	11	11	10	10	10	10
VN	5	4	5	5	6	5	6	6	3	3	3	3	4	3	4	4
r_s	—	0.99	1.00	1.00	—	0.98	1.00	1.00	—	0.97	1.00	1.00	—	0.90	1.00	1.00
r_w	—	0.99	1.00	1.00	—	0.99	1.00	1.00	—	0.98	1.00	1.00	—	0.89	1.00	1.00
WS	—	0.99	1.00	1.00	—	0.99	1.00	1.00	—	1.00	1.00	1.00	—	0.95	1.00	1.00

Note: r_s = Spearman coefficient; r_w = weighted rank measure of correlation; *WS* = rank similarity coefficient.

Table 10
Comparison of rankings by different aggregating methods in four years under **MinMax-Entropy**.

Country	2018				2015				2013				2009			
	MABAC	TOPSIS	RSR	VIKOR	MABAC	TOPSIS	RSR	VIKOR	MABAC	TOPSIS	RSR	VIKOR	MABAC	TOPSIS	RSR	VIKOR
BN	2	2	2	2	1	1	1	1	2	2	2	2	3	4	3	3
ID	4	4	4	4	4	5	4	4	5	5	5	5	5	6	5	5
KH	9	9	9	9	10	11	10	10	9	9	9	9	11	11	11	11
LA	11	11	11	11	8	8	8	8	8	8	8	8	9	8	9	9
MM	8	8	8	8	11	10	11	11	11	11	11	11	6	5	6	6
MY	3	1	3	3	3	2	3	3	4	4	4	4	2	2	2	2
PH	7	7	7	7	6	6	6	6	6	6	6	6	7	7	7	7
SG	1	3	1	1	2	3	2	2	1	1	1	1	1	1	1	1
TH	6	5	6	6	7	7	7	7	7	7	7	7	8	9	8	8
TL	10	10	10	10	9	9	9	9	10	10	10	10	10	10	10	10
VN	5	6	5	5	5	4	5	5	3	3	3	3	4	3	4	4
r_s	—	0.95	1.00	1.00	—	0.97	1.00	1.00	—	1.00	1.00	1.00	—	0.97	1.00	1.00
r_w	—	0.93	1.00	1.00	—	0.97	1.00	1.00	—	1.00	1.00	1.00	—	0.97	1.00	1.00
WS	—	0.86	1.00	1.00	—	0.94	1.00	1.00	—	1.00	1.00	1.00	—	0.97	1.00	1.00

Note: r_s = Spearman coefficient; r_w = weighted rank measure of correlation; *WS* = rank similarity coefficient.

Table 11
Grouping results by different analyses (2018).

Country	PCA	HCA	k1s_mean	**k4s_mean**	k20s_mean	k35s_mean	k72s_mean	Final group
SG	1	1	1	1	1	1	1	1
BN	1	1	1	1	1	1	1	1
ID	2	2	1	2	1	1	2	2
MY	2	2	1	2	1	2	2	2
VN	2	2	1	2	1	2	2	2
PH	2	2	2	2	2	2	2	2
TH	2	2	2	2	2	2	2	2
MM	3	3	2	3	2	2	2	3
KH	3	3	2	3	3	2	2	3
TL	3	3	3	3	3	3	3	3
LA	3	3	3	3	3	3	3	3

Table 12
Experiment outcomes from the k-means with Monte Carlo simulation (2018).

Results	Occurrence	Frequency	Probability (1% significance)	Probability (5% significance)	Probability (10% significance)
k1s_mean	302	0.302	0.269	0.279	0.284
k4s_mean	597	0.597	0.560	0.571	0.577
k20s_mean	44	0.044	0.031	0.035	0.036
k35s_mean	43	0.043	0.030	0.034	0.036
k72s_mean	14	0.014	0.008	0.097	0.010
Sum	1000	1	—	—	—

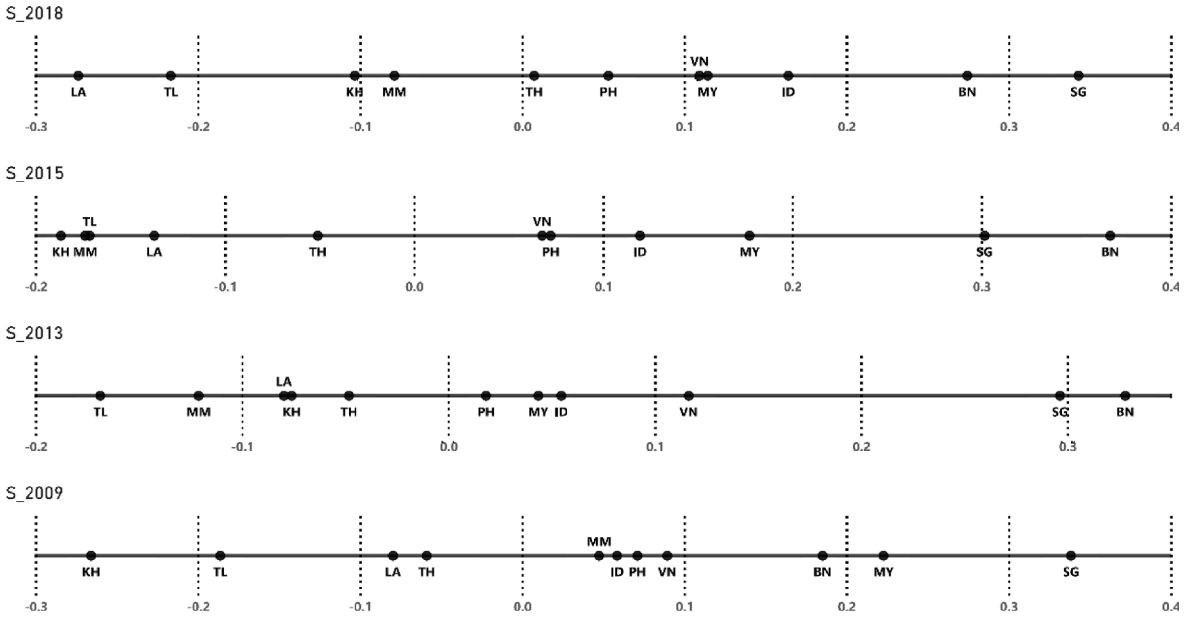


Fig. 5. Scatter distribution of S values from MABAC.

grouping result from our selected k4_mean is almost consistent with the PCA results, demonstrating the reliability of our k-means algorithm with Monte Carlo simulation. The scatter results of the PCA are presented in Fig. 6. We also introduce an HCA with centroid linkage as another robust comparison. Table 11 reports the HCA results for the countries in 2018, which are consistent with the selected k4s_mean. A dendrogram of the HCA is shown in Fig. 7.

We generate various grouping results, despite running the same k-means algorithm (program), because of the different initial centroid selections. As we do not specifically set the initial centroid, the initial center location is randomly chosen by the software, and a total of four types of grouping results emerge among these different selections of the initial centroid. Such inconsistency complicates the determination of trustworthy results, especially in small-sample analyses where uncertainty in one or two alternatives' grouping results may lead to large errors. Monte Carlo simulations effectively resolve this problem.

We also conduct the k-means Monte Carlo simulation on the country grouping for the other three years and used PCA grouping and

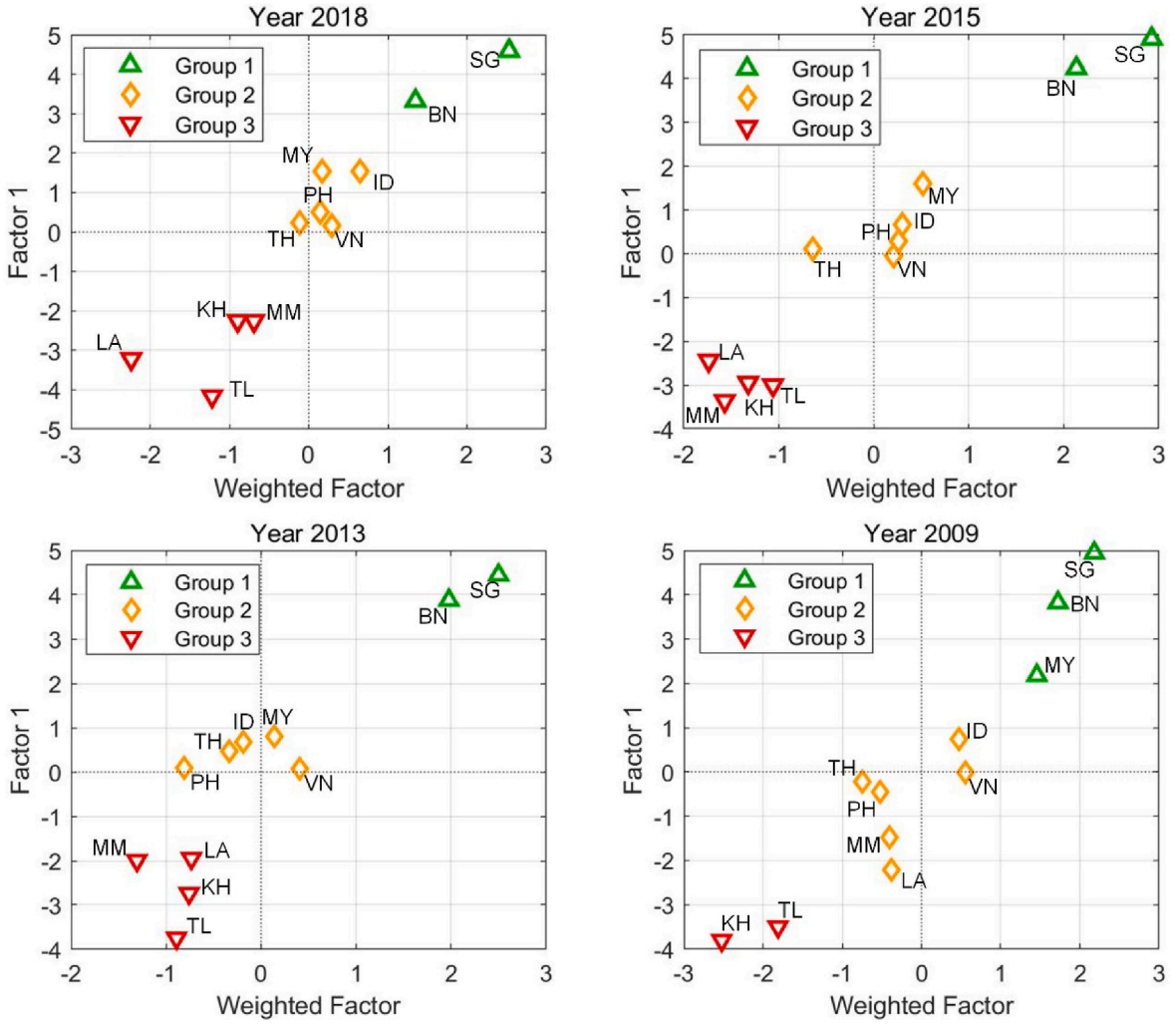


Fig. 6. Grouping from PCA for the four years.

HCA for comparison (PCA scatters and the HCA dendrogram are presented in Fig. 6 and Fig. 7). Table 13, Table 15, and Table 17 present the grouping results for 2015, 2013, and 2009, respectively, while Table 14, Table 16, and Table 18 present the corresponding frequency (probability) distributions of the different grouping results generated by Monte Carlo simulations. In addition, Fig. 5 shows the scatter of S values from MABAC of the 11 countries for verification. We select k1s_mean for 2015, k1s_mean for 2013, and k6s_mean for 2009 as the final results for these years because of their highest frequencies (probabilities) in the 1,000 simulations. The k-means results for 2015 and 2013 are highly consistent with the PCA results, except for the classification of "TH" in 2015, which is classified as "Intermediate" by the PCA but as "Poor Group" by k-means. The selected k-means results for 2015 and 2013 are also fully consistent with those of HCA. Among various grouping results generated by k-means for 2009, k1s_mean and k6s_mean are the most likely. These are explicitly reflected in Fig. 5, where the last four countries ("KH," "TL," "LA," and "TH") have ambiguous groupings—k1s_mean clusters these four countries as the "Poor Group," whereas the k6s_mean only clusters the last two countries ("KH" and "TL") as the "Poor Group" and allocates the others ("LA" and "TH") to the "Intermediate Group." When compared to PCA results (Table 17), both k1s_mean and k6s_mean show consistency. This is because the k1s_mean result is identical to the PCA result, while the division between k6s_mean and PCA only differs in the categorizing of "LA" and "TH." Similar results are obtained when comparing the k-means results with the HCA results, as the k1s_mean result is fully consistent with the HCA result, and the k6s-means result is most consistent with the HCA result, except the peripheral alternatives "LA" and "TH." Therefore, both results can be considered final. However, as the decision-making principle is to select the result with the highest frequency (probability), k6s_mean is eventually selected.

6. Practical implications

The application of our proposed MCDM methodology to road safety development of the ASEAN region delivers important implications for policy and practice. Our grouping process successfully divides the 11 countries into three grades based on their road safety

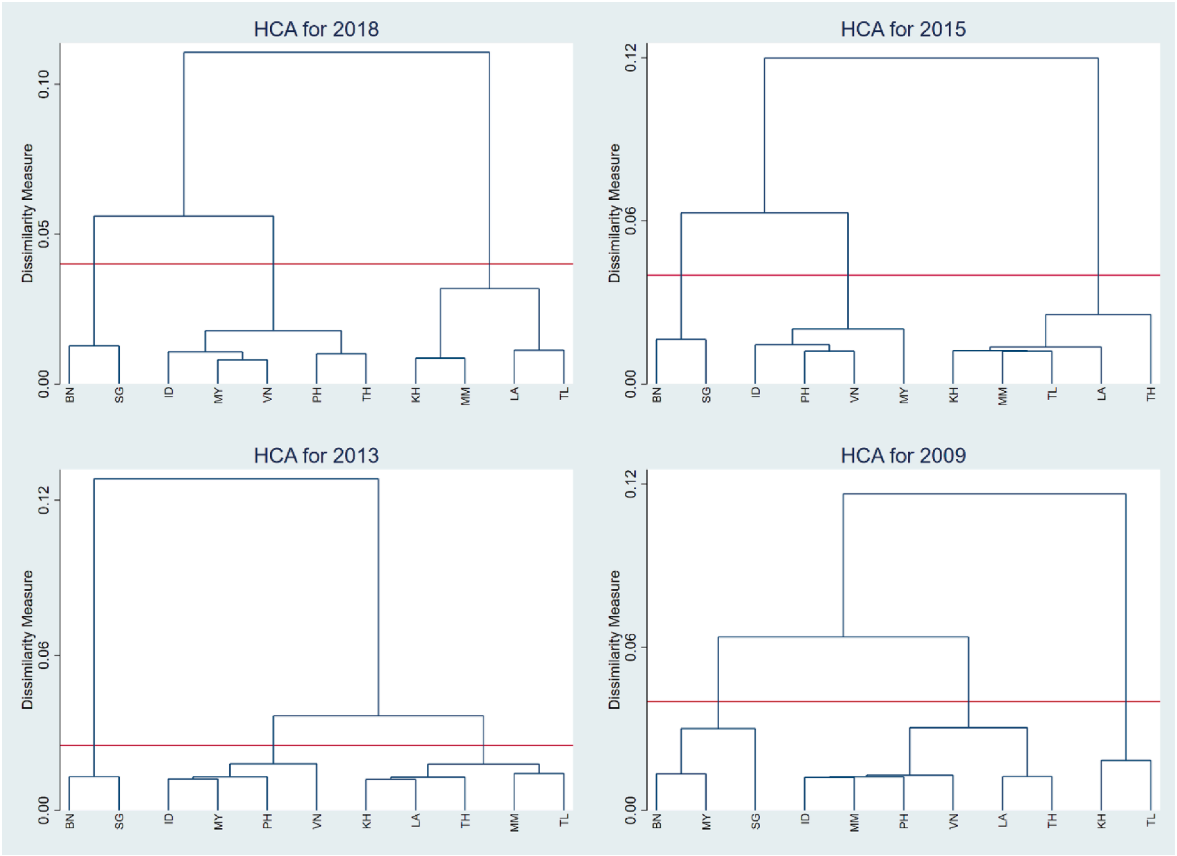


Fig. 7. Dendrogram of HCA for the four years.

Table 13
Grouping results by different analyses (2015).

Country	PCA	HCA	**k1s_mean**	k5s_mean	k8s_mean	Final group
BN	1	1	1	1	1	1
SG	1	1	1	1	1	1
MY	2	2	2	1	1	2
ID	2	2	2	1	2	2
PH	2	2	2	1	2	2
VN	2	2	2	1	2	2
TH	2	3	3	2	2	3
LA	3	3	3	3	2	3
TL	3	3	3	3	3	3
MM	3	3	3	3	3	3
KH	3	3	3	3	3	3

Table 14
Experiment outcomes from the k-means with Monte Carlo simulation (2015).

Results	Occurrences	Frequencies	Probability at 1% significant level	Probability at 5% significant level	Probability at 10% significant level
k1s_mean	523	0.523	0.486	0.497	0.503
k5s_mean	311	0.311	0.278	0.287	0.293
k8s_mean	166	0.166	0.140	0.148	0.151
Sum	1000	1	—	—	—

development (i.e., “Excellent,” “Intermediate,” and “Poor”). Within each group, the underperforming countries can improve by following the steps adopted by the benchmark country, which is the highest-ranked country in each group. Additionally, the road safety performance of individual indicators among ASEAN members within each cluster is deconstructed. The different clusters of

Table 15
Grouping results by different analyses (2013).

Country	PCA	HCA	**k1s_mean**	k8s_mean	Final group
BN	1	1	1	1	1
SG	1	1	1	2	1
VN	2	2	2	3	2
ID	2	2	2	3	2
MY	2	2	2	3	2
PH	2	2	2	3	2
TH	2	3	3	3	3
KH	3	3	3	3	3
LA	3	3	3	3	3
MM	3	3	3	3	3
TL	3	3	3	3	3

Table 16
Experiment outcomes from the k-means with Monte Carlo simulation (2013).

Results	Occurrences	Frequencies	Probability at 1% significant level	Probability at 5% significant level	Probability at 10% significant level
k1s_mean	968	0.968	0.952	0.958	0.960
k8s_mean	32	0.032	0.021	0.024	0.026
Sum	1000	1	—	—	—

Table 17
Grouping results by different analyses (2009).

Country	PCA	HCA	**k1s_mean**	k2s_mean	k4s_mean	**k6s_mean**	k11s_mean	k13s_mean	k48s_mean	Final group
SG	1	1	1	1	1	1	1	1	1	1
MY	1	1	1	2	1	1	1	1	1	1
BN	1	1	1	2	1	1	1	2	1	1
VN	2	2	2	2	1	2	1	2	2	2
PH	2	2	2	2	1	2	1	2	2	2
ID	2	2	2	2	1	2	1	2	2	2
MM	2	2	2	2	1	2	1	2	2	3
TH	2	2	2	3	2	3	2	3	2	3
LA	2	2	2	3	2	3	2	3	3	3
TL	3	3	3	3	2	3	3	3	3	3
KH	3	3	3	3	3	3	3	3	3	3

Table 18
Experiment outcomes from the k-means with Monte Carlo simulation (2009).

Results	Occurrences	Frequencies	Probability at 1% significant level	Probability at 5% significant level	Probability at 10% significant level
k1s_mean	222	0.222	0.193	0.201	0.206
k2s_mean	124	0.124	0.102	0.108	0.111
k4s_mean	53	0.053	0.039	0.043	0.045
k6s_mean	298	0.298	0.266	0.275	0.280
k11s_mean	149	0.149	0.125	0.131	0.135
k13s_mean	112	0.112	0.091	0.097	0.100
k48s_mean	42	0.042	0.030	0.033	0.035
Sum	1000	1	—	—	—

countries in the four years are presented in a visualized graph in Fig. 8.

The deconstruction of the 18 indicators using 2018 as an example is illustrated in Fig. 9. For instance, in the “Intermediate” group, ID has the highest rank and can be deemed as the benchmark. However, being the benchmark does not necessarily indicate absolute perfection. As we can observe, ID may exhibit an overall advantage, but it does not surpass other members in the same class in every aspect. Although it performs better in indicators concerning the enforcement scores of road safety laws (i.e., C_{41}–C_{43}), which may provide positive references for other countries, it lags in indicators related to personal traffic risks (A_{21}–A_{31}). Moreover, all the countries in this group are at a disadvantage in terms of macroeconomic development, considering indicators such as gross domestic product per capita (C_{32}) and citizen life expectancy (C_{33}). Laggard countries should compare their safety performance relative to the corresponding quantitative gap of each indicator and learn from the experience of countries that have performed well on relevant indicators to match the benchmark. However, blindly emulating the benchmark may not always be appropriate. Countries should consider their unique characteristics and local conditions to adjust suitably. In summary, our deconstruction process provides a

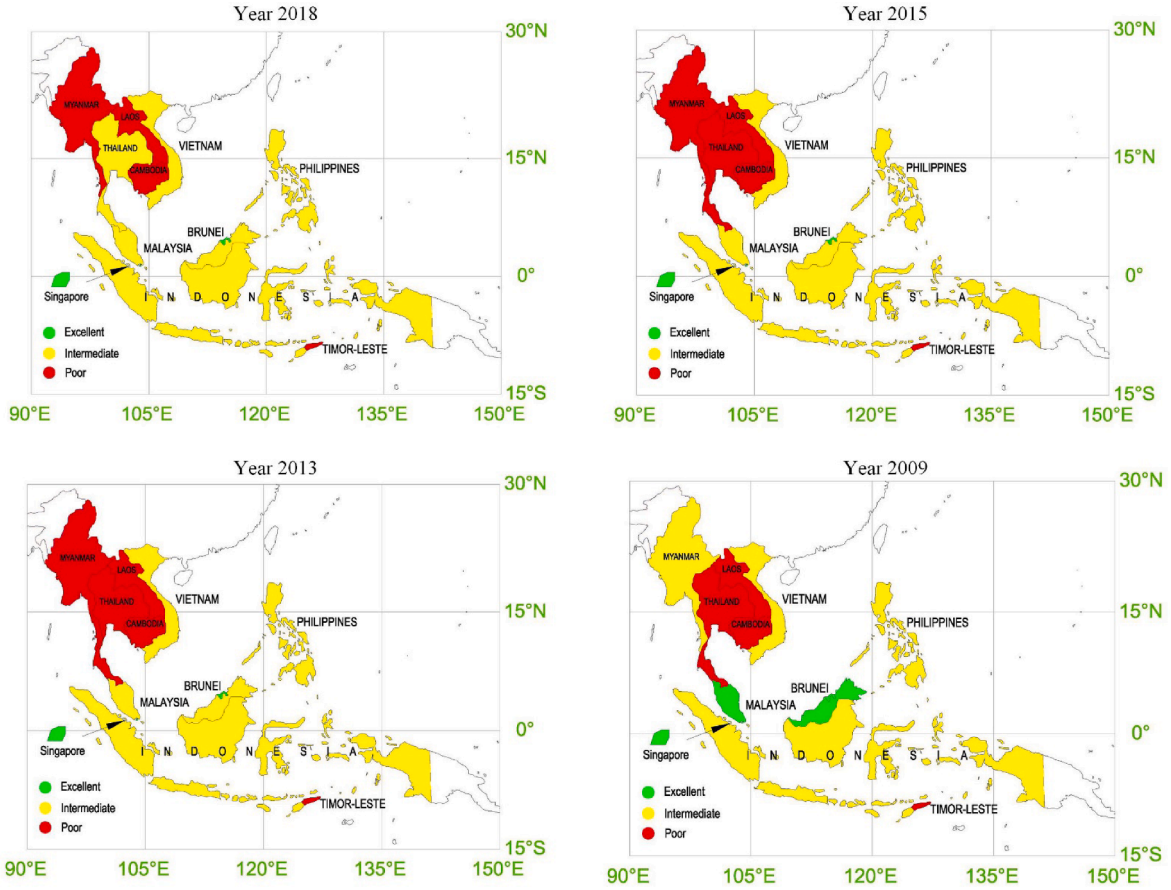


Fig. 8. Geographical distribution of ASEAN members regarding transport safety accomplishment.

comparison of road safety performance between countries, helping policymakers identify the areas in which their countries excel and areas that require improvement. This information enables policymakers to implement targeted measures to improve road safety in their respective countries.

7. Conclusion

Selecting an appropriate MCDM method to obtain a credible decision remains a challenge for DMs. This is due to the inherent complexity, sensitivity, and uncertainty involved in the decision-making process. Therefore, this study developed a hybrid MCDM model with Monte Carlo simulation (i.e., CRITIC–MABAC–Kmeans with Monte Carlo simulation) that integrates weighting, aggregating, and grouping into a one-step framework. The model exhibits considerable stability, reliability, and adaptability and aims to address MCDM problems that arise from compounded sensitivity and model-related uncertainty, especially in scenarios involving small sample sizes.

Within the practice of MABAC in the case of the ASEAN region, we conduct a comparative study of different normalization, weighting, and aggregating methods, focusing on two types of stability: internal and external. We find that MABAC is generally less susceptible to disruption by changes in indicator weights than other aggregating methods, such as TOPSIS, RSR, and VIKOR. When VIKOR, RSR, and TOPSIS are used as controls, MABAC results show extremely high consistency. This suggests that MABAC is a reliable tool that can be applied to many complex MCDM scenarios. In addition, as the k-means method is used for grouping in this study, the selection of the initial centroid can significantly affect the final grouping decision for our small-sample case. Therefore, a Monte Carlo simulation is employed to address the inconsistency and mitigate small-sample bias. The simulation in the k-means algorithm reveals that one method to solve the difficulty of choosing an initial centroid is not to choose it but to let the software randomly determine the starting point and run the algorithm multiple times. DMs can then select a reliable final grouping result based on that with the highest probability of occurrence.

The proposed hybrid MCDM model offers two benefits. First, it enriches the existing decision-making system, thereby expanding the breadth and depth of the available approaches. The results of this study show that our hybrid model is a stable, reliable, and applicable instrument for rational and reasonable decision-making. Given the stability (consistency) of the MABAC solution, the MABAC method can be adapted to many complicated decision-making problems to tackle uncertain decision-making issues,

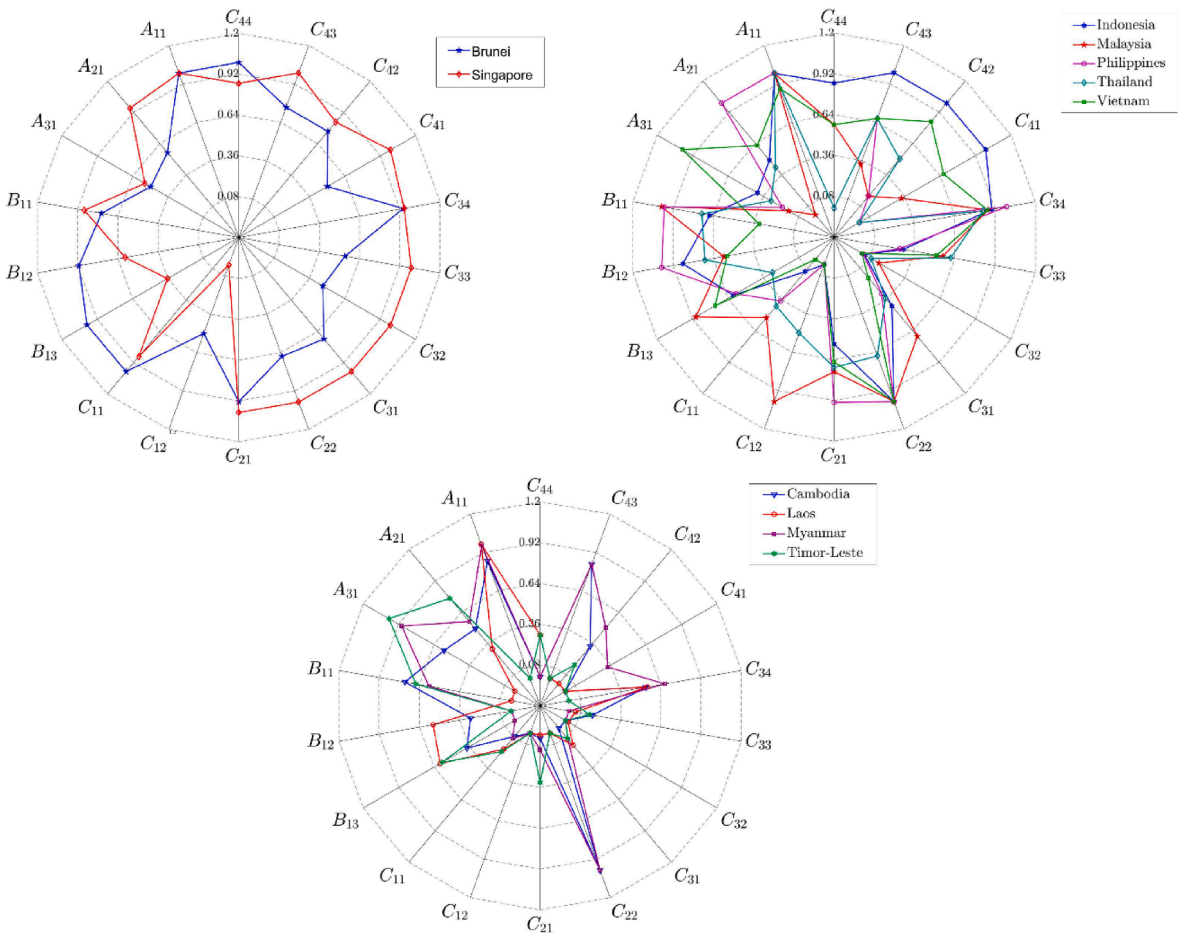


Fig. 9. Deconstruction of transport safety accomplishment concerning individual indicators among the ASEAN members within each cluster. (*Note*: The data used for the above comparison has been normalized by the MinMax method).

particularly for some important cases, such as benchmarking road safety development, in which valuable policy implications rely on trustworthy decision-making. The successful implementation of Monte Carlo simulation-nested k-means clustering may serve as a precursor to utilizing the k-means algorithm in small-sample cases and effectively solving the inconsistency problem instigated by an ambiguous initial centroid. Indeed, it extends the practical range of the k-means algorithm and allows k-means clustering to be more effective for small-sample cases. Second, it provides a strategic direction and decision basis for industrial systems, such as road safety development. Our hybrid MCDM model provides a benchmark for assessing road safety performance. These findings could encourage the implementation of effective actions and measures, promote best practices, and inspire political leaders to establish a transport system with optimal safety standards.

The limitations of our proposed hybrid MCDM model lie in two aspects. First, the model ignores DMs' subjective preferences, particularly the weighting method. Specifically, the model utilizes an objective method (CRITIC) that uses the objective attributes of data for scientific evaluation and reduces the uncertainty from subjective epistemes. However, it is important to note that in certain situations, such as decision-making related to socioeconomic development, DMs' subjective preferences may play a significant role in assigning importance to criteria and making decisions based on the current state of societal development. Second, the proposed model neglects DMs' risk preference because it uses a deterministic approach to evaluate alternatives. It assumes that DMs make rational choices based solely on the information provided without considering their personal attitudes toward risk.

In terms of future research directions, implementing a more advanced weighting method in the proposed hybrid MCDM model is feasible. For instance, it may be worthwhile exploring the use of subjective weighting that can be adapted to individual preferences and opinion-based evaluations using the proposed MCDM model. Additionally, innovative mixed-weight methods that integrate both objective and subjective weights can be developed. As MABAC was proven less predisposed to the disturbance of weight methods in this study, the flexible hybrids of MABAC to other weight methods for different purposes could produce reliable outcomes. Moreover, there can be a potential improvement in the current MABAC methodology, which involves incorporating parameters to measure DMs' risk preferences. In addition, in future studies, generating bootstrapping samples by empirical distribution in combination with the Monte Carlo simulation proposed in this study can be considered as another approach to freeing k-means clustering methods from the

threat of small-sample bias and the problem of divergent clustering results for peripheral individuals owing to inconsistent initial centroids.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

The authors acknowledge financial, capacity-building, and technological support from Harvard-China Project on Energy, Economy and Environment at Harvard University. The authors would like to thank the Editor-in-Chief (Prof. Witold Pedrycz) and three anonymous reviewers for their valuable comments and constructive suggestions, which have been of great help in improving the quality of this paper.

Appendix A

(1) Spearman coefficient.

The most frequently used approach is the Spearman coefficient (r_s) [46–48] suggested in the following expression: Spearman's coefficient was interpreted as the proportion of variability in the ranks of one variable that could be explained by the other variable.

$$r_s = 1 - \frac{6 \cdot \sum_{i=1}^{n} (A_{xi} - A_{yi})^2}{n \cdot (n^2 - 1)}$$

where A_{xi} is the ranking of alternative i in aggregating method x, A_{yi} is the ranking of alternative i in comparative aggregating method y, and n is the number of alternatives in the MCDM case.

(2) Weighted rank measure of correlation.

Another typical coefficient for depicting ranking similarity is the weighted rank measure of correlation (r_w) [49], which employs a linear function of two ranks to assign weights to the distance between them. It accounts for the position at which the difference in ranking occurs, and is expressed by the following formula.

$$r_w = 1 - \frac{6 \cdot \sum_{i=1}^{n} (A_{xi} - A_{yi})^2 \cdot ((n - A_{xi} + 1) + (n - A_{yi} + 1))}{n^4 + n^3 - n^2 - n}$$

(3) Rank similarity coefficient *WS*.

There is a new coefficient of ranking similarity (*WS*) in decision-making problems that tends to be robust and sensitive to changes in ranking [50]. The formula used to calculate *WS* is as follows.

$$WS = 1 - \sum_{i=1}^{N} (2^{-2A_{xi}} \cdot \frac{|A_{xi} - A_{yi}|}{\max\{|1 - A_{xi}|, |N - A_{xi}|\}})$$

where N is the length of the ranking and A_{xi} and A_{yi} refers to the rank values from the two aggregating methods for comparison. The weight of a given comparison is determined based on the significance of the position in the ranking.

All three coefficients (r_s, r_w, and *WS*) reflect the similarity of the ranking results obtained by different MCDM methods; the closer their absolute value is to 1, the more similar the ranking results are in the two MCDM methods.

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